# Euclid's Elements — Book XI
# Proposition 11

Formal Reconstruction — Technical Documentation

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*From a given elevated point to draw a straight line perpendicular to a given plane.*

## 1 Introduction

Proposition XI.11 is the first Book XI construction in the present reconstruction whose target is the existence of a perpendicular from an arbitrary point outside a plane. Given a plane $\pi$ and an elevated point $A \notin \pi$, the theorem constructs a line through $A$ that is perpendicular to $\pi$.

The proposition is a natural test of the architecture developed in XI.4–XI.8. Its final statement is spatial, but the construction repeatedly moves into ordinary planar Hilbert geometry on explicitly chosen plane slices. In the nontrivial branch three planes are relevant:

$$\pi, \qquad \sigma = \text{plane through } A \text{ and } BC, \qquad \tau = \text{plane through } AD \text{ and } DE.$$

The proof also exposes an important distinction between the theorem proved and the proof route used. The current production proof follows Euclid's construction closely and therefore invokes XI.8. Since XI.8 uses the plane-local Euclidean parallel axiom, the public Lean theorem assumes `HilbertSpaceEuclidean`. This records the strength of the current proof, not a claim that the existence statement itself is axiomatically minimal.

## 2 Euclid's Statement

Euclid's Proposition XI.11 states:

> From a given elevated point to draw a straight line perpendicular to a given plane.

Let $A$ be the elevated point and let $\pi$ be the plane of reference. The required construction is a line $AF$ through $A$ such that

$$AF \perp \pi.$$

## 3 Classical Configuration

Choose a straight line $BC$ at random in the plane of reference. Draw $AD$ from $A$ perpendicular to $BC$. If $AD$ is already perpendicular to the reference plane, the construction is complete.

Otherwise draw $DE$ in the reference plane through $D$, with

$$DE \perp BC.$$

Then draw $AF$ from $A$ perpendicular to $DE$, and through $F$ draw $GH$ parallel to $BC$.

Thus the reference plane contains

$$BC, \qquad DE, \qquad GH,$$

while the two lines

$$AD, \qquad AF$$

leave the reference plane and meet it at $D$ and $F$, respectively.

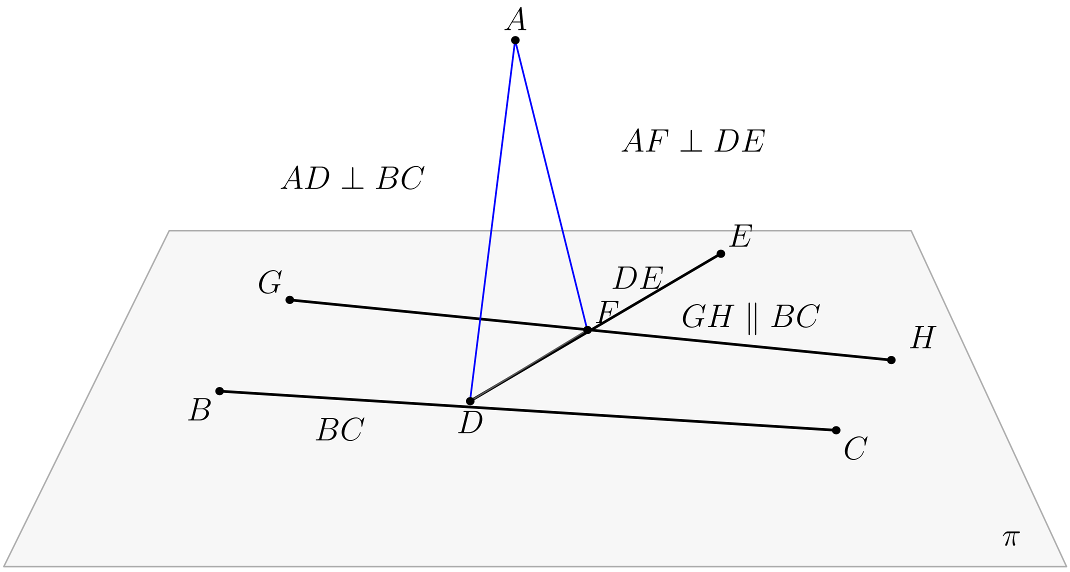


Figure 1: The classical configuration of Proposition XI.11. Black lines lie in the reference plane $\pi$; blue segments leave $\pi$. The construction starts with $AD \perp BC$, then uses $DE \perp BC$, $AF \perp DE$, and $GH \parallel BC$. The desired conclusion is $AF \perp \pi$.

## 4 Classical Proof

The classical proof is short but structurally dense.

First, choose $BC \subset \pi$ and draw

$$AD \perp BC$$

from the elevated point $A$ by Proposition I.12. Euclid immediately splits into two cases.

If $AD \perp \pi$, the required perpendicular has already been drawn.

Assume therefore that $AD$ is not perpendicular to $\pi$. In the reference plane draw

$$DE \perp BC$$

through $D$ by Proposition I.11. Next draw

$$AF \perp DE$$

from $A$ by a second application of I.12. Finally draw through $F$

$$GH \parallel BC$$

by I.31.

Now $BC$ is perpendicular to both $DA$ and $DE$. These two lines intersect at $D$, so Proposition XI.4 gives

$$BC \perp \text{plane}(ED, DA).$$

Since $GH \parallel BC$, Proposition XI.8 gives

$$GH \perp \text{plane}(ED, DA).$$

By XI.Def.3, a line perpendicular to a plane is perpendicular to every line of that plane through the foot. The line $AF$ belongs to plane$(ED, DA)$ and meets $GH$ at $F$. Hence

$$GH \perp AF,$$

and therefore also

$$AF \perp GH.$$

But $AF \perp DE$ by construction. Thus $AF$ is perpendicular at $F$ to the two intersecting lines $DE$ and $GH$ of the reference plane. A second application of XI.4 yields

$$AF \perp \text{plane}(DE, GH).$$

The plane through $DE$ and $GH$ is the plane of reference, hence

$$AF \perp \pi.$$

The local classical dependency chain is therefore

$$
\begin{gathered}
A \notin \pi, \quad BC \subset \pi \\
\Downarrow \\
\mathrm{I.12} : AD \perp BC \\
\Downarrow \\
AD \perp \pi \quad \text{or} \quad AD \not\perp \pi \\
\Downarrow \\
\mathrm{I.11} : DE \perp BC \\
\Downarrow \\
\mathrm{I.12} : AF \perp DE \\
\Downarrow \\
\mathrm{I.31} : GH \parallel BC \\
\Downarrow \\
\mathrm{XI.4} : BC \perp \mathrm{plane}(ED, DA) \\
\Downarrow \\
\mathrm{XI.8} : GH \perp \mathrm{plane}(ED, DA) \\
\Downarrow \\
\mathrm{XI.Def.3} : GH \perp AF \\
\Downarrow \\
AF \perp DE, \quad AF \perp GH \\
\Downarrow \\
\mathrm{XI.4} : AF \perp \pi.
\end{gathered}
$$

No use of XI.9 or XI.10 occurs in this classical chain. The Book XI dependencies needed locally are XI.4 and XI.8.

## 5 What is genuinely three-dimensional here?

The proof is not simply a planar perpendicular construction repeated in space. The essential issue is selecting the correct plane for each planar construction and then transporting its conclusion back to ambient space.

The first auxiliary plane is

$$\sigma = \mathrm{plane}(A, BC).$$

It exists because $A \notin BC$. Inside $\sigma$, the point $A$ and the line $BC$ are ordinary planar objects, so I.12 can be applied in the induced geometry $\mathrm{PlaneGeo}(\sigma)$.

After the foot $D$ is obtained, the line $DE$ is constructed inside the reference plane $\pi$.

The second auxiliary plane is

$$\tau = \mathrm{plane}(AD, DE).$$

The lines $AD$ and $DE$ intersect at $D$ and are distinct. Inside $\mathrm{PlaneGeo}(\tau)$, the second I.12 construction produces the line $AF \perp DE$.

The three plane slices have a particularly transparent intersection pattern:

$$\pi \cap \sigma = BC, \qquad \sigma \cap \tau = AD, \qquad \pi \cap \tau = DE,$$

at the level of the carriers used in the construction.

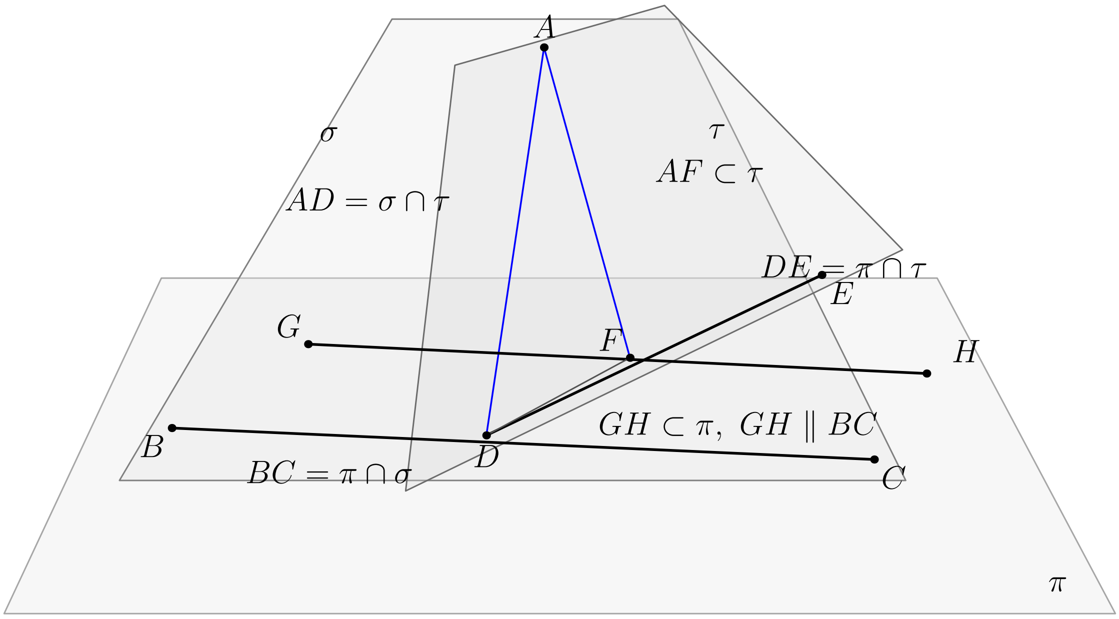


Figure 2: The three plane slices used by the reconstruction. The reference plane $\pi$ contains $BC, DE, GH$. The plane $\sigma$ carries the first planar perpendicular construction and shares $BC$ with $\pi$. The plane $\tau$ carries $AD, DE, AF$; it shares $AD$ with $\sigma$ and $DE$ with $\pi$. The final argument transfers perpendicularity from $BC$ to $GH$ through XI.8 and then applies XI.4 in $\pi$.

The figure should not be interpreted as a coordinate model. It is only a perspective rendering of incidence and plane membership. The Lean proof uses no coordinates, vectors, scalar products, normals, or numerical angle measure.

## 6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_11
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    [HSE : HilbertSpaceEuclidean Geo]
    (pi : S.Plane)
    (A : Geo.Point)
    (hApi : Not (S.OnPlane A pi)) :
    exists l F,
      H.OnLine A l /\
      HilbertLinePerpendicularPlaneAt Geo l pi F
```

The conclusion returns both the perpendicular carrier $l$ and its foot $F$. It asserts that $A$ lies on $l$ and that $l$ is perpendicular to $\pi$ at $F$.

The predicate

```
HilbertLinePerpendicularPlaneAt Geo l pi F
```

is the spatial target relation established for XI.4. It expresses XI.Def.3 synthetically: $F$ lies on the line and in the plane, and the line is perpendicular to every line of the plane through $F$.

# 7 Spatial / Planar Architecture Used

## 7.1 Ambient spatial layer

The ambient construction uses

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertLineInPlane
HilbertSpaceLinesParallel
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

There is no global ambient instance

```
HilbertOrder Geo
```

and no global ambient instance

```
HilbertCongruence Geo
```

installed merely to make planar theorems available in space.

## 7.2 Induced planar geometry

Planar constructions are performed in explicitly named plane slices using

```
PlanePoint
PlaneLine
PlaneGeo
```

The proof uses three instances of induced planar geometry:

$$\text{PlaneGeo}(\sigma), \qquad \text{PlaneGeo}(\pi), \qquad \text{PlaneGeo}(\tau).$$

The first I.12 construction is performed in PlaneGeo($\sigma$). The construction of $DE$ and the I.31

parallel construction are performed in PlaneGeo($\pi$). The second I.12 construction is performed in PlaneGeo($\tau$).

### 7.3 Plane–space bridges

The principal bridges used by the proof include

```
planeGeo_rightAngle_iff_ambient
planeGeo_linesDisjoint_iff_ambient
HilbertSpaceIncidence.line_in_plane
HilbertLinePerpendicularPlaneAt.incidence
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

The proof also uses the ambient symmetry normalizer

```
hilbert_space_linesPerpendicularAt_symm
```

when XI.4 requires the perpendicularity relation in the opposite carrier orientation.

## 8 Proof Architecture of the Reconstruction

The formal proof follows the classical route but expands the incidence and plane-selection steps that the printed diagram leaves implicit.

### 8.1 1. Choosing the reference line

The proof first obtains three noncollinear points of the reference plane:

```
hilbert_three_noncollinear_on_plane
```

Two of them, $B, C$, determine the line $bc$. The theorem

```
HilbertSpaceIncidence.line_in_plane
```

then proves

$$bc \subset \pi.$$

Since $A \notin \pi$, it follows immediately that

$$A \notin bc.$$

## 8.2 2. Constructing the first auxiliary plane

The helper

```
hilbert_plane_through_line_and_external_point
```

constructs the plane $\sigma$ containing $bc$ and $A$.

This step supplies exactly the missing incidence premise needed before a planar I.12 theorem can be applied. The classical phrase "draw $AD$ from $A$ perpendicular to $BC$" presupposes that $A$ and $BC$ have first been put in a common plane.

## 8.3 3. First I.12 construction in $\sigma$

Inside

$$\mathrm{PlaneGeo}(\sigma)$$

the theorem

```
hilbert_perpendicular_from_point_exists
```

constructs the foot $D$ on $bc$ and gives a right-angle witness. The ambient carrier $ad$ through $D, A$ is then constructed explicitly and proved to lie in $\sigma$.

The right-angle data are packaged as

```
HilbertLinesPerpendicularAt Geo bc ad D
```

using the bridge

```
planeGeo_rightAngle_iff_ambient
```

and explicit noncollinearity.

## 8.4 4. Euclid's case split

The formal proof uses the explicit split

```
by_cases hADpi :
    HilbertLinePerpendicularPlaneAt Geo ad pi D
```

If this proposition holds, the proof returns $ad, D$ immediately.

This is not a Lean-specific mathematical invention. It is exactly Euclid's sentence: if $AD$ is also perpendicular to the plane of reference, the construction is already complete.

### 8.5 5. Constructing $DE$ in the reference plane

In the nontrivial branch the point $D$ is known to lie in $\pi$. The formal proof constructs opposite points on $bc$ in PlaneGeo($\pi$) and then uses

```
hilbert_right_angle_exists_nondegenerate
```

to obtain a nondegenerate right-angle direction through $D$. The resulting carrier $de$ satisfies

$$de \subset \pi, \qquad bc \perp de \text{ at } D.$$

Thus the formal proof realizes the role of Euclid I.11 without importing a separate I.11 proposition wrapper in the production file.

### 8.6 6. Constructing the second auxiliary plane

The proof establishes

$$ad \neq de.$$

Indeed, if $ad = de$, then $A$ would lie on $de \subset \pi$, contrary to $A \notin \pi$.

The theorem

```
hilbert_plane_through_two_intersecting_lines
```

then constructs the plane $\tau$ containing $ad$ and $de$.

### 8.7 7. Second I.12 construction in $\tau$

Inside

$$\text{PlaneGeo}(\tau)$$

the proof applies

```
hilbert_perpendicular_from_point_exists
```

a second time, now from $A$ to $de$. This yields the foot $F$ and the carrier $af$, with

$$af \subset \tau, \qquad de \perp af \text{ at } F.$$

Because $de \subset \pi$, the point $F$ also lies in the reference plane.

## 8.8 8. The hidden fact $F \notin BC$

Before I.31 can be applied through $F$, the formal proof must establish that $F$ does not lie on $bc$.

Assume $F \in bc$. Since $D, F$ lie on both $bc$ and $de$, line uniqueness forces $F = D$. The two lines $ad$ and $af$ then contain the two distinct points $D, A$, so line uniqueness forces

$$ad = af.$$

The proof already knows

$$ad \perp bc, \qquad ad \perp de$$

at $D$. Since $bc, de \subset \pi$, XI.4 would imply

$$ad \perp \pi,$$

contradicting the nontrivial branch assumption.

Therefore

$$F \notin bc.$$

This is one of the main pieces of hidden formal data in XI.11.

## 8.9 9. I.31 in the reference plane

Now $B, C, F$ are noncollinear in PlaneGeo($\pi$). The proof applies

```
euclid_proposition_31
```

to obtain a point $G$ and a line $gh$ through $F, G$ parallel to $bc$ in the induced plane.

The result of I.31 is initially planar point-pair parallelism. The proof converts it to disjointness of the actual carriers using

```
hilbert_mem_pointLine_iff_onLine
```

and then transports planar disjointness back to ambient space by

```
planeGeo_linesDisjoint_iff_ambient
```

Finally it packages

```
HilbertSpaceLinesParallel Geo bc gh
```

with the reference plane $\pi$ as the explicit common carrier plane.

### 8.10 10. First XI.4: $BC \perp \tau$

In the plane $\tau$, the lines $ad$ and $de$ are distinct and meet at $D$. The proof has

$$bc \perp ad, \qquad bc \perp de.$$

Therefore

```
euclid_proposition_11_4
```

gives

$$bc \perp \tau.$$

This is the first use of XI.4 in the construction.

### 8.11 11. XI.8: transfer to the parallel $GH$

The proof now has

$$bc \parallel gh, \qquad bc \perp \tau.$$

It applies

```
euclid_proposition_11_8
```

and obtains a point $K$ such that

$$gh \perp \tau \text{ at } K.$$

This is exactly the classical XI.8 step, but the formal theorem returns the foot existentially. Euclid's diagram already identifies the relevant intersection as $F$; Lean requires that identification to be proved.

### 8.12 12. Normalizing the XI.8 foot: $K = F$

The proof knows from the XI.8 conclusion that

$$K \in gh, \qquad K \in \tau,$$

and from the construction that

$$F \in gh, \qquad F \in \tau.$$

Suppose $K \neq F$. Since the two distinct points $K, F$ lie on $gh$ and in $\tau$, spatial incidence gives

$$gh \subset \tau.$$

But $gh \perp \tau$ at $K$. Applying the universal line–plane-perpendicularity condition to the line $gh$ itself would give

$$gh \perp gh.$$

The theorem

```
hilbert_linesPerpendicularAt_ne
```

then yields the absurd conclusion $gh \neq gh$.

Hence

$$K = F.$$

This argument is purely synthetic and uses the nondegeneracy already built into `HilbertLinesPerpendicularAt`.

### 8.13 13. XI.Def.3 gives $AF \perp GH$

After replacing $K$ by $F$, the proof has

$$gh \perp \tau \text{ at } F.$$

Since

$$af \subset \tau, \qquad F \in af,$$

the universal line condition is unpacked by

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

to obtain

$$gh \perp af.$$

The symmetry helper

```
hilbert_space_linesPerpendicularAt_symm
```

normalizes this to

$$af \perp gh.$$

The same helper normalizes the previously constructed $de \perp af$ to

$$af \perp de.$$

### 8.14 14. Proving $DE \neq GH$

The final XI.4 application requires two distinct lines of the reference plane. This distinctness is not left to the diagram.

Suppose

$$de = gh.$$

Since $D \in de$, it follows that $D \in gh$. But $D \in bc$, while the spatial parallelism package $bc \parallel gh$ contains disjointness of these two ambient carriers. Contradiction.

Thus

$$de \neq gh.$$

### 8.15 15. Final XI.4 in the reference plane

Both $de$ and $gh$ lie in $\pi$, meet at $F$, and are distinct. The proof has

$$af \perp de, \qquad af \perp gh.$$

A final call to

```
euclid_proposition_11_4
```

therefore gives

$$af \perp \pi \text{ at } F.$$

The returned witnesses are $af$ and $F$.

## 9 Lean Formalization

The production module imports

```
import CGJteamLab.Proposition11_8
import CGJteamLab.Proposition12
import CGJteamLab.Proposition31
```

The direct imports therefore expose XI.8, the Book I perpendicular-from- external-point construction used as I.12, and I.31.

XI.4 is invoked twice by the proof and is available transitively through the established Book XI import chain. The production theorem does not import or invoke XI.9 or XI.10.

The principal reusable declarations consumed by XI.11 are:

```
hilbert_three_noncollinear_on_plane
hilbert_plane_through_line_and_external_point
hilbert_perpendicular_from_point_exists
planeGeo_rightAngle_iff_ambient
planeGeo_opposite_points_on_line_congruent_to
hilbert_right_angle_exists_nondegenerate
hilbert_plane_through_two_intersecting_lines
hilbert_space_linesPerpendicularAt_symm
euclid_proposition_31
hilbert_mem_pointLine_iff_onLine
planeGeo_linesDisjoint_iff_ambient
euclid_proposition_11_4
euclid_proposition_11_8
HilbertLinePerpendicularPlaneAt.incidence
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
hilbert_linesPerpendicularAt_ne
```

No XI.11-specific helper theorem is exported. The full construction is contained in the proof of `euclid_proposition_11_11`.

# 10 Hidden Formal Data

The classical diagram suppresses several obligations that are explicit in the Lean proof.

## 10.1 The first I.12 needs a plane

The statement $A \notin \pi$ does not by itself put $A$ and the chosen line $BC \subset \pi$ in a common planar geometry. The plane $\sigma$ through $A$ and $BC$ must be constructed before planar I.12 can be invoked.

## 10.2 Every constructed line needs a carrier proof

After constructing ambient lines $ad$, $de$, $af$, and $gh$, the proof records explicitly which plane contains each line. These are the data that permit later calls to XI.4, XI.8, and induced-plane theorems.

## 10.3 The two feet are not automatically the same

XI.8 returns a foot $K$ of $gh$ on $\tau$. The earlier construction already supplied the point $F \in gh \cap \tau$. The equality $K = F$ is a real incidence theorem, not a definitional simplification.

### 10.4 Distinctness for the final XI.4

The statement $de \neq gh$ follows from the disjointness part of $bc \parallel gh$, but it must be proved explicitly before the two lines can be packaged as distinct `PlaneLines` of $\pi$.

### 10.5 Orientation of perpendicularity

The project relation `HilbertLinesPerpendicularAt` carries ordered line arguments. Several classical sentences silently use symmetry of perpendicularity. The proof normalizes these orientations explicitly with `hilbert_space_linesPerpendicularAt_symm`.

## 11 Relationship with Earlier Book XI Propositions

The only earlier Book XI propositions invoked by the mathematical proof are XI.4 and XI.8.

XI.4 is used twice:

$$bc \perp ad, \quad bc \perp de \Longrightarrow bc \perp \tau,$$

and later

$$af \perp de, \quad af \perp gh \Longrightarrow af \perp \pi.$$

XI.8 is used once:

$$bc \parallel gh, \quad bc \perp \tau \Longrightarrow gh \perp \tau.$$

The source audit is important here because the current working sequence skips XI.9 and XI.10. Heath's proof of XI.11 does not cite either proposition. Nor does the production Lean source import or invoke them. They are therefore genuinely absent from both the classical local DAG and the actual Lean DAG for XI.11.

XI.7 is also not a local dependency of XI.11. The spatial parallelism used here is created directly by I.31 inside the reference plane and then packaged as `HilbertSpaceLinesParallel`.

## 12 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical Book I dependencies are:

$$\text{I.12}, \quad \text{I.11}, \quad \text{I.12}, \quad \text{I.31}.$$

The formal proof calls the established I.12 theorem

```
hilbert_perpendicular_from_point_exists
```

twice, once in $\sigma$ and once in $\tau$.

The role of I.11 is reconstructed from the lower-level right-angle construction interface in PlaneGeo($\pi$). The proof does not import a separate proposition-I.11 wrapper.

The I.31 step is represented directly by

```
euclid_proposition_31
```

inside PlaneGeo($\pi$).

Book Zero enters transitively through the established planar Hilbert construction and right-angle infrastructure. XI.11 introduces no new Book Zero declaration.

The ambient three-space remains separated from planar order and congruence. All ordinary planar constructions are carried out in a named `PlaneGeo`; the spatial classes provide only the incidence, congruence transport, and plane-local Euclidean data required to connect those slices.

## 13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_11

#print axioms Geometry.euclid_proposition_11_11
```

and Lean reports:

```
'Geometry.euclid_proposition_11_11' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

This is the same inherited global axiom profile already accepted at XI.8. There is no new XI.11-specific global axiom constant and no `sorry`/`admit`.

The explicit theorem signature nevertheless contains

```
[HSE : HilbertSpaceEuclidean Geo]
```

because the proof invokes XI.8. Thus Group IV is a genuine hypothesis of the current production theorem.

This must not be confused with a claim of logical necessity for the target statement. The present theorem records an Euclid-faithful construction through XI.8. The source comparison in the Borsuk–Szmielew appendix shows that a stronger, axiomatically more economical existence-and-uniqueness theorem is available in a pre-Euclidean congruence development. Therefore a future neutral strengthening of the Lean result is a legitimate separate project question, not part of the present XI.11 production milestone.

At this checkpoint:

- new proposition-specific geometric axiom: no;
- new spatial axiom declaration: no;
- Euclidean parallel axiom / Hilbert Group IV: used, via XI.8;
- new continuity assumption: no;
- new proposition-local helper theorem: no;
- new reusable `Hilbert3DInterface` theorem: no;
- new `Hilbert3DAxioms` declaration: no;
- new `PlaneGeo` bridge theorem: no;
- global ambient `HilbertOrderGeo`: not installed;
- global ambient `HilbertCongruenceGeo`: not installed;
- public theorem compiles: yes, as reported at this checkpoint;
- full repository build: clean, as reported at this checkpoint;
- axiom audit: completed;
- `sorry`/`admit`: no.

## 14 Dependency Summary

The classical local chain is

$$
\begin{array}{c}
A \notin \pi, \quad BC \subset \pi \\
\downarrow \\
\text{I.12} \\
\downarrow \\
AD \perp BC \\
\downarrow \\
\text{case } AD \perp \pi \text{ or not} \\
\downarrow \\
\text{I.11}, \text{I.12}, \text{I.31} \\
\downarrow \\
DE \perp BC, \quad AF \perp DE, \quad GH \parallel BC \\
\downarrow \\
\text{XI.4} \\
\downarrow \\
BC \perp \text{plane}(ED, DA) \\
\downarrow \\
\text{XI.8} \\
\downarrow \\
GH \perp \text{plane}(ED, DA) \\
\downarrow \\
\text{XI.Def.3} \\
\downarrow \\
AF \perp GH \\
\downarrow \\
\text{XI.4} \\
\downarrow \\
AF \perp \pi.
\end{array}
$$

The actual formal chain is

$$
\begin{gathered}
\text{Hilbert spatial incidence/order/congruence} \\
\downarrow \\
\texttt{HilbertSpaceEuclidean}\ \text{(Group IV)} \\
\downarrow \\
bc \subset \pi, \quad A \notin bc \\
\downarrow \\
\sigma = \text{plane}(A, bc) \\
\downarrow \\
\text{I.12 in } \text{PlaneGeo}(\sigma) \\
\downarrow \\
bc \perp ad \\
\downarrow \\
\texttt{by\_cases}\ ad \perp \pi \\
\downarrow \\
de \subset \pi, \quad bc \perp de \\
\downarrow \\
\tau = \text{plane}(ad, de) \\
\downarrow \\
\text{I.12 in } \text{PlaneGeo}(\tau) \\
\downarrow \\
de \perp af, \quad F \in \pi \\
\downarrow \\
F \notin bc \\
\downarrow \\
\text{I.31 in } \text{PlaneGeo}(\pi) \\
\downarrow \\
\texttt{HilbertSpaceLinesParallel}\ bc\, gh \\
\downarrow \\
\text{XI.4 in } \tau \\
\downarrow \\
bc \perp \tau \\
\downarrow \\
\text{XI.8} \\
\downarrow \\
gh \perp \tau \text{ at } K \\
\downarrow \\
K = F \\
\downarrow \\
\text{XI.Def.3} : af \perp gh \\
\downarrow \\
de \neq gh \\
\downarrow \\
\text{XI.4 in } \pi \\
\downarrow \\
\texttt{euclid\_proposition\_11\_11}.
\end{gathered}
$$

The dependency classification is:

| | |
|---|---|
| New proposition-specific global axiom | no |
| New spatial axiom class | no |
| Continuity | no |
| Group IV / Euclidean parallelism | yes, via XI.8 |
| Direct production imports | XI.8, I.12 module, I.31 |
| Classical explicit Book XI dependencies | XI.4, XI.8 |
| XI.9 / XI.10 | not used |
| New proposition-local helper theorem | no |
| New PlaneGeo bridge | no |
| Coordinates / vectors / scalar product | no |
| `sorry/admit` | no |

## 15 Conclusion

Proposition XI.11 is a compact classical construction whose formal reconstruction exercises almost the entire plane–space interface developed so far. The key mathematical idea remains Euclid's: construct two successive planar perpendiculars, introduce a parallel in the reference plane, transfer a line–plane perpendicularity by XI.8, and close with XI.4.

Lean makes the spatial bookkeeping visible. The first I.12 needs the plane $\sigma$; the second needs the plane $\tau$; the I.31 construction must be converted from planar parallelism to ambient spatial parallelism; the foot returned by XI.8 must be proved equal to $F$; and the final two reference-plane carriers must be proved distinct.

No analytic geometry enters. The production theorem is fully synthetic, compiles in the current project, and has the inherited axiom audit

$$[\texttt{propext}, \texttt{Classical.choice}, \texttt{Geometry.bookZero\_nullSegment1}, \texttt{Quot.sound}].$$

Its present Group IV hypothesis is inherited from the Euclid-faithful use of XI.8. The independent source comparison shows that reduction of this hypothesis is a possible later strengthening, not an unresolved defect in the completed XI.11 milestone.